

THE UNIVERSITY OF
SYDNEY

Tutorial Class	
Student Name	
SID	

CONFIDENTIAL QUIZ PAPER

This paper should not be removed from the quiz venue.

COMP5339 – Data Engineering

Tutorial Quiz – 1

Individual Quiz (5%)

Quiz Writing Time: **30 minutes**
Friday, 17 April, at 06 PM

Instructions to Students

- Please ensure you correctly complete your **tutorial** and **Student ID** details.
- This is a **closed-book** test. **Electronic devices** (including phones, laptops, tablets, etc.) are **not permitted**. Please place all devices in your bag and keep your bag on the floor.
- This tutorial quiz consists of **3 multiple-choice questions** and three short descriptive questions. **All questions must be answered.**
- Write all answers in the spaces provided on this paper.
- Questions carry **different marks**. The mark value for each question is shown at the start of the question. The **total mark** for this quiz is **10**, which contributes to 5% of the total marks.
- For short-answer questions, please write **legibly**. Record your final answers in **ink**. **Do not** use a pencil or **red ink**.

Please submit the completed quiz paper to your tutor **before leaving the room**.

Part I – Multiple Choice Questions

Question 1: This question has three parts. Tick the boxes corresponding to each correct answer. Each question carries one mark.

A. Which statement best describes the role of staging in an OLTP → OLAP pipeline?

- ☐ Staging tables are the final tables used by BI dashboards; fact/dim tables are just temporary
- ☐ Staging replaces the OLTP system by serving user transactions with high concurrency
- ☐ Staging exists only to store aggregated results (e.g., monthly revenue)
- ☐ **Staging stores a raw or lightly cleaned copy of source data to support reruns, deduplication, and incremental loading before building fact/dim tables**

B. You can either (i) run SQL that filters/aggregates in the DB and return a small result set, or (ii) bulk-load the whole table into Pandas and process it there. Which is the strongest technical reason for choosing option (i)?

- ☐ Pandas cannot do joins or group-bys
- ☐ SQL cannot express aggregations compared to Pandas
- ☐ Bulk-loading is always slower, even for small datasets
- ☐ **DBMS can optimise queries, and bulk-loading requires the full dataset to fit into memory**

C. A data engineering team is ingesting semi-structured data from multiple sources where fields vary across records. They decide to use a schema-late approach. Which of the following is the main advantage of this decision?

- ☐ Enforces strict consistency across all records
- ☐ Reduces storage size compared to relational databases
- ☐ Eliminates the need for data validation entirely
- ☐ **Allows ingestion of heterogeneous data without a predefined schema**

Part II – Short Answer Questions

Provide answers in the provided boxes.

Scenario 1: The Table below shows order lines for one customer (note that such orders exist with several customers). Two dashboards (developed by two different teams) show different “**monthly revenue**” numbers. One includes only order status with “Shipped”; the other includes “Cancelled” and “On Hold”. Both teams claim they are correct.

ORDERNUMBER	CUSTOMERNAME	ORDERDATE	STATUS	SALES
20001	Australian Collectors, Co.	1/10/2004	Shipped	3600
20015	Australian Collectors, Co.	1/18/2004	Cancelled	1204
20022	Australian Collectors, Co.	1/25/2004	On Hold	870

Answer the following:

How do you enforce consistent metric definitions across the organisation, and what pipeline components help prevent or detect these inconsistencies (e.g., checks, documentation, semantic layer, etc.)? **(2 Marks)**

Define metrics centrally (data catalogue/semantic layer) with agreed filters (e.g., only status='Shipped'). Publish tables and prevent ad-hoc redefinition by creating views. Add data tests (for accepted statuses and revenue formula checks) and documentation.

Scenario 2: A fact table grows to billions of rows. Most queries filter by date and region, but performance and cost are rising.

Question 2: What strategies would you use to improve performance and control cost (e.g., partitioning, aggregates), and how do you decide which to implement? **(2 Marks)**

- If most queries filter by time, then partition by date first.
- If queries also filter heavily by region, then cluster/sort on those.
- If the same aggregations power dashboards repeatedly, then materialised views/marts.

Scenario 3: Your analysts want near-real-time dashboards. The OLTP database is already under heavy load. Running large analytical queries against it slows down customer operations.

Question 3: What extraction approach would you choose (see options below), and how would you architect the system so analytical workloads do not impact OLTP performance? Choose one choice below and justify your selection. **(3 Mark)**

1. Change Data Capture (CDC), i.e., as soon as the data changes
2. Replication
3. Scheduled extracts

1. I would avoid querying the OLTP database directly for dashboards because analytical scans and joins can compete with transactional workloads (locks, CPU, I/O). Instead, I would use CDC (reading the database's transaction log) to stream changes into an analytics environment.
2. CDC reads changes from the transaction log rather than running large joins/aggregations on OLTP tables.
3. OLTP already writes to logs for recovery; CDC just consumes those logs.